# YIFEI SUN

Staff Software Engineer · Platform Infrastructure & Experimentation · 13 Years

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## SUMMARY

Staff-level platform engineer with 13 years building internal developer infrastructure, experimentation systems, and large-scale distributed services in Java. Built an experimentation and evaluation platform with A/B testing, parameter search, and automated analysis. Led cross-team redesign of a high-traffic API serving millions of daily requests (p99 250ms→50ms, availability 98%→99.9%), validating the migration at 50% production traffic through parity testing and controlled rollout. Established architecture standards and observability practices adopted across 10+ teams and 3 departments. Combines deep hands-on Java engineering with cross-organizational technical leadership and mentoring.

## LEADERSHIP HIGHLIGHTS

- Built experimentation platform with A/B testing, parameter search, and automated evaluation pipelines
- Drove architecture and observability standards across 10+ teams, 3 departments via RFCs and design docs
- Led cross-team API redesign validated via controlled rollout at 50% traffic: p99 5× reduction, 99.9% availability
- Mentored senior engineers to Principal-level scope; built shared platform services and internal developer tooling

**Technologies:** Java, Spring Boot, GraphQL (Netflix DGS), Python, Node.js · AWS, Kubernetes, Kafka, Redis, PostgreSQL · Microservices, Event-Driven, CQRS · A/B Testing, Experimentation · Datadog, Grafana, OpenTelemetry, Distributed Tracing · CI/CD, Docker

## INDEPENDENT PROJECT

### Experimentation & Evaluation Platform

- Built an end-to-end experimentation system for decision evaluation: A/B simulation, offline backtesting, parameter search (grid & Bayesian optimization), and automated analysis with configurable metrics and thresholds.
- Designed a configurable policy framework with DB-backed configs, environment-specific overrides, and a scheduler-driven tuning loop that sweeps parameters and publishes promotable config versions — enabling safe, data-driven iteration.
- Architected a hybrid rules + multi-path inference engine over heterogeneous data sources with full observability, telemetry collection, and auditability across the entire experiment lifecycle.

## EXPERIENCE

**CarGurus — Principal Software Engineer / Technical Lead** Oct 2019 – Present

### High-Traffic API Modernization — Experiment-Driven Migration

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core API behind CarGurus' most-visited page (~8,000 rpm, millions of daily impressions); validated the new implementation at 50% production traffic via controlled A/B experiment.
- Introduced a new orchestration service and GraphQL boundary, consolidating 12 downstream integrations into 5 well-defined services, reducing fan-out and simplifying failure modes.
- Built a parity-testing framework that automatically compared new vs. legacy responses across full payload and data-integrity dimensions, enabling safe, data-driven rollout decisions.

  **p99: 250 ms → 50 ms** (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)
- Defined architectural standards and observability requirements for new microservices; instrumented services with structured logging, distributed tracing, and SLO-based latency budgets per downstream dependency.

### GraphQL Federation — Delivery Lookup DGS Subgraph

- Designed and integrated a Netflix DGS subgraph into CarGurus' federated GraphQL gateway, exposing vehicle delivery data to frontend, backend, and AI consumers from a single graph.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies to avoid JIT compilation latency spikes on cold starts.

### ChatGPT Apps SDK MCP Server (0→1)

- Sole backend engineer designing and implementing an MCP server compatible with the OpenAI Apps SDK, exposing CarGurus capabilities into ChatGPT as a high-visibility proof-of-concept.

- Delivered a performant implementation iterating on ambiguous requirements; designed for safe, incremental production rollout alongside existing APIs.

### Digital Shopping & Online Leads — Rapid Experimentation

- Led backend design for a new digital shopping product launched during the pandemic, using microservice and event-driven architecture for inventory, merchandising, and lead flows.
- Built infrastructure for rapid product experimentation — enabling the company to quickly launch, measure, and safely retire product bets without destabilizing core systems.

### Cross-Org Platform Leadership & Observability Standards

- Established patterns for API design, observability, caching, and distributed orchestration adopted across multiple teams; authored design docs, RFCs, and decision records driving consensus.
- Drove adoption of structured logging, distributed tracing, and metrics-driven development practices. Mentored senior engineers on system design and path to Principal-level scope.

## Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

### Authentication & Identity Modernization

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, unifying auth across internal systems, an enterprise partner platform, and a large external community. Introduced social login and mobile login; migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

### Event Sourcing & CQRS Initiative

- Drove early design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

### Cross-Org Collaboration

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations with clear observability and rollback paths.

## Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

### Monolith → Microservices Decomposition

- Among the earliest engineers during hypergrowth (first customer to hundreds in a few years); designed and implemented core backend services (Order Service, Labor Service, foundational APIs) that became the basis for future growth.
- Built initial AWS deployment pipelines before a dedicated DevOps team existed; created the first automated testing framework before dedicated QA, enabling safer releases during rapid product expansion.

### Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners (e.g., workforce management), ensuring reliable bidirectional sync and schema evolution. Partnered with PMs, sales, and support to translate early customer feedback into platform capabilities.

## The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

# EDUCATION

**M.S. in Electrical and Computer Engineering — Northeastern University, Boston** 2010 – 2013

**B.S. in Information Science — Beijing University of Aeronautics and Astronautics** 2006 – 2010